Team members (1-3): Thomas Chen, Janys Li
Project title: Multi-modality model for motion drift realignment in biomechanics modeling.
Preliminary hypothesis: The IMU-based motion capture drift can be mitigated by context-specific constraints and auxiliary sensor modalities such as light, pressure, magnetometer, and audio cues. Signals from multiple IMU can be integrated to estimate motion efficiency and provide improvement suggestions in sports.
Category of project: biomechanics study, device prototype

Topic(s) from course related to project that will be incorporated (must check at least 1):
- Simple motion models
- Biomechanics musculoskeletal simulation / modeling
- Experimental biomechanics measurement tools
- Human-subject study related design

List 5 research papers related to this idea (search on google scholar):
1. Wittmann F, Lambercy O, Gassert R. Magnetometer-Based Drift Correction During Rest in IMU Arm Motion Tracking. Sensors (Basel). 2019 Mar 15;19(6):1312. doi: 10.3390/s19061312. PMID: 30884745; PMCID: PMC6471153.
2.Kim, D., Jin, Y., Cho, H. et al. Learning-based 3D human kinematics estimation using behavioral constraints from activity classification. Nat Commun 16, 3454 (2025). https://doi.org/10.1038/s41467-025-58624-6
3.Jin, Y., Zhou, Y.M., Pathak, P. et al. Minimal-calibration multimodal wearable sensing for long-duration three-dimensional shoulder kinematics. Nat. Sens. 1, 502–512 (2026). https://doi.org/10.1038/s44460-026-00059-7
4. Kok, M., Weygers, I., Osman, H., Weber, D., Li, R., Seel, T., & Seth, A. (2026). Inertial human motion capture: From biomechanics to recent sensor fusion methods and back. *Journal of Biomechanics*, *206*, 113542. https://doi.org/10.1016/j.jbiomech.2026.113542
5.Phan, V., Li, Z., Meinders, E. et al. Biomechanics-informed inertial tracking achieves the accuracy of marker-based kinematics. Nat Commun 17, 7927 (2026). https://doi.org/10.1038/s41467-026-75981-y
6. Mollyn, V., Arakawa, R., Goel, M., Harrison, C., & Ahuja, K. (2023). IMUPoser: Full-body pose estimation using IMUs in phones, watches, and earbuds. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Article 529, pp. 1–12). Association for Computing Machinery. https://doi.org/10.1145/3544548.3581392
7. Xu, V., Gao, C., Hoffmann, H., & Ahuja, K. (2024). MobilePoser: Real-time full-body pose estimation and 3D human translation from IMUs in mobile consumer devices. In Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology (Article 70, pp. 1–11). Association for Computing Machinery. https://doi.org/10.1145/3654777.3676461

Proposed project milestone 1: Implement an IMU-based motion capture prototype and collect preliminary data to test consistency with camera-tracking ground truth. Compare measurement consistency and precision against other motion capture methods.

Proposed project milestone 2: Develop alignment (sensor fusion) algorithms and apply motion pattern detection in example sports.